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INSTRUCTIONS

Set a timer of 3 Hrs and solve all 75 Questions. Then match with the answer key and calculate on the basis of +4, -1.

Once you are done with it refer video discussion to resolve doubts and comment your report In the comment section. Report includes total score and learnings

Last page of this file has answer key to all 75 Questions

CLICK FOR SOLUTION – PART 2



CLICK FOR PART 1

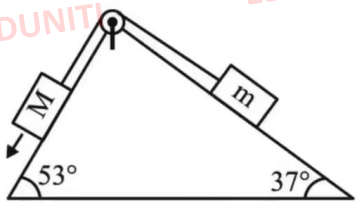
1. Time periods of oscillation of the same simple pendulum measured using four different measuring clocks were recorded as 4.62 s, 4.632 s, 4.6 s and 4.64 s. The arithmetic mean of these reading in correct significant figure is.

(1) 4.623 s (2) 4.62 s
(3) 4.6 s (4) 5 s

2. The diameter of a sphere is measured using a vernier caliper whose 9 divisions of main scale are equal to 10 divisions of vernier scale. The shortest division on the main scale is equal to 1 mm. The main scale reading is 2 cm and second division of vernier scale coincides with a division on main scale. If mass of the sphere is 8.635 g, the density of the sphere is:

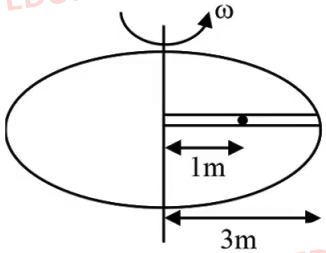
(1) 2.5 g/cm^3 (2) 1.7 g/cm^3
(3) 2.2 g/cm^3 (4) 2.0 g/cm^3

3. In the given arrangement of a doubly inclined plane two blocks of masses M and m are placed. The blocks are connected by a light string passing over an ideal pulley as shown. The coefficient of friction between the surface of the plane and the blocks is 0.25. The value of m, for which M = 10 kg will move down with an acceleration of 2 m/s^2, is : (take g = 10 m/s^2 and tan 37° = 3/4)



(1) 9 kg (2) 4.5 kg
(3) 6.5 kg (4) 2.25 kg

4. A circular table is rotating with an angular velocity of ω rad/s about its axis (see figure). There is a smooth groove along a radial direction on the table. A steel ball is gently placed at a distance of 1m on the groove. All the surface are smooth. If the radius of the table is 3 m, the radial velocity of the ball w.r.t. the table at the time ball leaves the table is $x\sqrt{2}\omega$ m/s, where the value of x is…….



5. A simple pendulum of length 1 m has a wooden bob of mass 1 kg. It is struck by a bullet of mass 10^{-2} kg moving with a speed of 2×10^2 ms^{-1}. The bullet gets embedded into the bob. The height to which the bob rises before swinging back is.
(use g = 10 m/s^2)

(1) 0.30 m (2) 0.20 m
(3) 0.35 m (4) 0.40 m

6. In a system two particles of masses m_1 = 3kg and m_2 = 2kg are placed at certain distance from each other. The particle of mass m_1 is moved towards the center of mass of the system through a distance 2cm. In order to keep the center of mass of the system at the original position, the particle of mass m_2 should move towards the center of mass by the distance ____ cm.

7. A metal wire of uniform mass density having length L and mass M is bent to form a semicircular arc and a particle of mass m is placed at the centre of the arc. The gravitational force on the particle by the wire is:

(1) $\frac{GMm\pi}{2L^2}$

(2) 0

(3) $\frac{GmM\pi^2}{L^2}$

(4) $\frac{2GmM\pi}{L^2}$

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8. To find the spring constant (k) of a spring experimentally, a student commits 2% positive error in the measurement of time and 1% negative error in measurement of mass. The percentage error in determining value of k is :

(1) 3% (2) 1%

(3) 4% (4) 5%

9. The density and breaking stress of a wire are 6×10^4 kg /m^3 and 1.2×10^8 N/m^2 respectively. The wire is suspended from a rigid support on a planet where acceleration due to gravity is $\frac{1^{rd}}{3}$ of the value on the surface of earth. The maximum length of the wire with breaking is ..... m (take, g = 10 m/s^2)

10. A tuning fork resonates with a sonometer wire of length 1 m stretched with a tension of 6 N. When the tension in the wire is changed to 54 N, the same tuning fork produces 12 beats per second with it. The frequency of the tuning fork is _____ Hz.

11. A parallel plate capacitor with width 4 cm, length 8 cm and separation between the plates of 4mm is connected to a battery of 20 V. A dielectric slab of dielectric constant 5 having length 1cm, width 4 cm and thickness 4 mm is inserted between the plates of parallel plate capacitor. The electrostatic energy of this system will be.....$\in_0$ J.

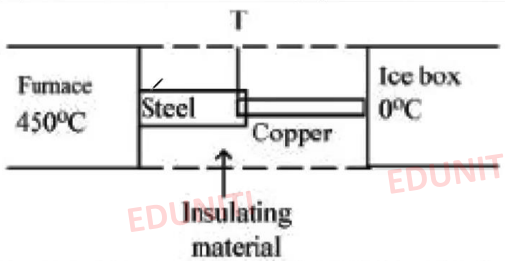
(Where $\in_0$ is the permittivity of free space)

12. If K_1 and K_2 are the thermal conductivities L_1 and L_2 are the lengths and A_1 and A_2 are the cross sectional areas of steel and copper rods respectively such that $\frac{K_2}{K_1} = 9, \frac{A_1}{A_2} = 2, \frac{L_1}{L_2} = 2$.

Then, for the arrangement as shown in the figure. The value of temperature T of the steel – copper junction in the steady state will be :

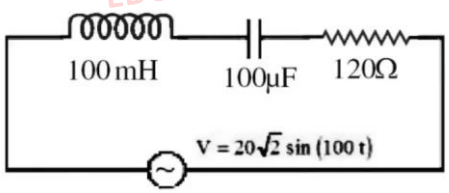
(A) 18°C (B) 14 °C

(C) 45 °C (D) 150 °C



13. In a coil of resistance 8Ω, the magnetic flux due to an external magnetic field varies with time as $\phi = \frac{2}{3}\left(9 - t^2\right)$. The value of total heat produced in the coil, till the flux becomes zero, will be_____J.

14. An AC source is connected to an inductance of 100 mH, a capacitance of 100 μ F and a resistance of 120 Ω as shown in figure. The time in which the resistance having a thermal capacity 2 J°/C will get heated by 16°C is _____s.



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15. A coil of inductance 1 H and resistance 100 Ω is connected to a battery of 6 V. Determine approximately :

(a) The time elapsed before the current acquires half of its steady – state value

(b) The energy stored in the magnetic field associated with the coil at an instant 15 ms after the circuit is switched on. (Given In2 = 0.693, $e^{-3/2} = 0.25$)

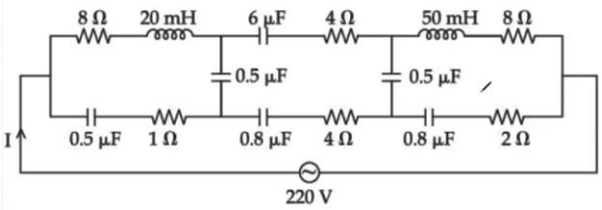
(A) t = 10 ms; U = 2 mJ

(B) t = 10 ms; U = 1 mJ

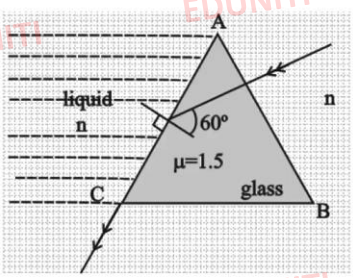
(C) t = 7 ms; U = 1 mJ

(D) t = 7 ms; U = 2 mJ

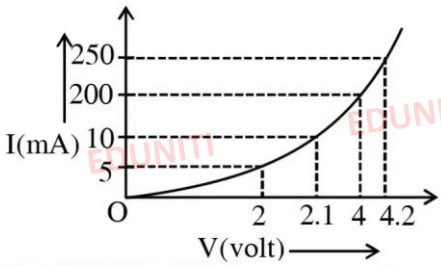
16. The effective current I in the given circuit at very high frequencies will be ____A



17. In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index 'n'. For incident angle 60° at the side AC, the refracted light beam just grazes along face AC. The refractive index of the liquid $n = \frac{\sqrt{x}}{4}$. The value of x is_____ .
(Given refractive index of glass = 1.5)



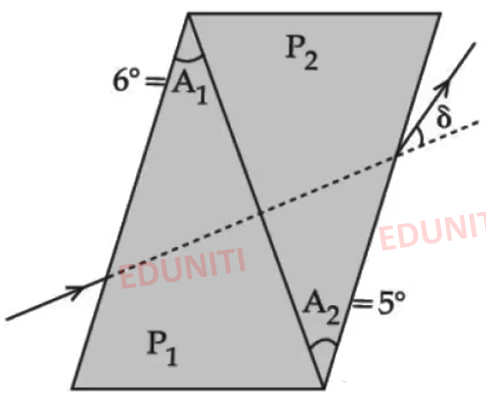
18. The I-V characteristics of a p-n junction diode in forward bias is shown in the figure. The ratio of dynamic resistance, corresponding to forward bias voltages of 2V and 4V respectively, is :



(A) 1 : 2 (C) 1 : 40

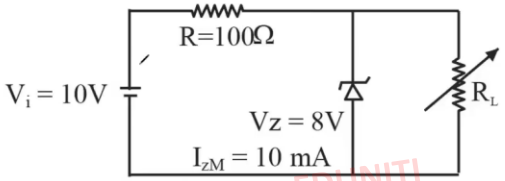
(B) 5 : 1 (D) 20 : 1

19. A thin prism of angle 6° and refractive index for yellow light (n_Y)1.5 is combined with another prism of angle 5° and n_Y = 1.55. The combination produces no dispersion. The net average deviation (δ) produced by the combination is $\left(\frac{1}{x}\right)^{\circ}$. The value of x is.....



CLICK FOR SOLUTION

20. A Zener of breakdown voltage $V_Z = 8V$ and maximum zener current, $I_{ZM} = 10$ mA is subjected to an input voltage $V_i = 10V$ with series resistance $R = 100\Omega$. In the given circuit R_L represents the variable load resistance. The ratio of maximum and minimum value of R_L is _____



21. Sun light falls normally on a surface of area 36 cm^2 and exerts an average force of 7.2×10^{-9} N within a time period of 20 minutes. Considering a case of complete absorption, the energy flux of incident light is

(A) 25.92×10^{2} W/cm^2
(B) 8.64×10^{-6} W/cm^2
(C) 6.0 W/cm^2
(D) 0.06 W/cm^2

22. Consider a rectangular sheet of solid material of length ℓ = 9 cm and width d = 4 cm. The coefficient of linear expansion is $\alpha = 3.1 \times 10^{-5}$ K^{-1} at room temperature and one atmospheric pressure. The mass of sheet m = 0.1 kg and the specific heat capacity $C_v = 900$ J $kg^{-1}K^{-1}$. If the amount of heat supplied to the material is 8.1×10^2 J then change in area of the rectangular sheet is :-

(1) 2.0×10^{-6} m^2 (2) 3.0×10^{-7} m^2
(3) 6.0×10^{-7} m^2 (4) 4.0×10^{-7} m^2

23. The centre of mass of a thin rectangular plate (fig - x) with sides of length a and b, whose mass per unit area (σ) varies as $\sigma = \frac{\sigma_0 x}{ab}$ (where σ_0 is a constant), would be_____

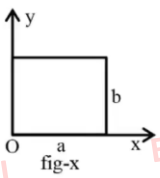
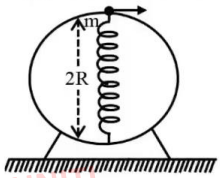


fig-x

(1) $\left(\frac{2}{3}a, \frac{b}{2}\right)$ (2) $\left(\frac{2}{3}a, \frac{2}{3}b\right)$
(3) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (4) $\left(\frac{1}{3}a, \frac{b}{2}\right)$

24. A bead of mass 'm' slides without friction on the wall of a vertical circular hoop of radius 'R' as shown in figure. The bead moves under the combined action of gravity and a massless spring (k) attached to the bottom of the hoop. The equilibrium length of the spring is 'R'. If the bead is released from top of the hoop with (negligible) zero initial speed, velocity of bead, when the length of spring becomes 'R', would be (spring constant is 'k', g is acceleration due to gravity)



(1) $2\sqrt{gR + \frac{kR^2}{m}}$ (2) $\sqrt{2Rg + \frac{4kR^2}{m}}$
(3) $\sqrt{2Rg + \frac{kR^2}{m}}$ (4) $\sqrt{3Rg + \frac{kR^2}{m}}$

25. A steel wire of length 2 m and Young's modulus 2.0×10^{11} Nm^{-2} is stretched by a force. If Poisson ratio and transverse strain for the wire are 0.2 and 10^{-3} respectively, then the elastic potential energy density of the wire is ____ $\times 10^5$ (in SI units)

CLICK FOR SOLUTION

26. In the experiment for measurement of viscosity 'η' of given liquid with a ball having radius R, consider following statements.

A. Graph between terminal velocity V and R will be a parabola

B. The terminal velocities of different diameter balls are constant for a given liquid.

C. Measurement of terminal velocity is dependent on the temperature.

D. This experiment can be utilized to assess the density of a given liquid.

E. If balls are dropped with some initial speed, the value of η will change.

Choose the correct answer from the options given below:

(1) B, D and E only

(2) A, C and D only

(3) C, D and E only

(4) A, B and E only

27. Two slabs with square cross section of different materials (1, 2) with equal sides (l) and thickness d_1 and d_2 such that $d_2 = 2d_1$ and $l > d_2$. Considering lower edges of these slabs are fixed to the floor, we apply equal shearing force on the narrow faces. The angle of deformation is $\theta_2 = 2\theta_1$. If the shear moduli of material 1 is 4×10^9 N/m^2, then shear moduli of material 2 is $x \times 10^9$ N/m^2, where value of x is ____.

28. Consider a completely full cylindrical water tank of height 1.6 m and cross-sectional area 0.5 m^2. It has a small hole in its side at a height 90 cm from the bottom. Assume, the cross-sectional area of the hole to be negligibly small as compared to that of the water tank. If a load 50 kg is applied at the top surface of the water in the tank then the velocity of the water coming out at the instant when the hole is opened is : ($g = 10$ m/s^2) 3-1

(1) 3 m/s (3) 2 m/s

(2) 5 m/s (4) 4 m/s

CLICK FOR SOLUTION

29. Two simple pendulums having lengths l_1 and l_2 with negligible string mass undergo angular displacements θ_1 and θ_2, from their mean positions, respectively. If the angular accelerations of both pendulums are same, then which expression is correct ?

(1) $\theta_1 l_2^2 = \theta_2 l_1^2$ (2) $\theta_1 l_1 = \theta_2 l_2$

(3) $\theta_1 l_1^2 = \theta_2 l_2^2$ (4) $\theta_1 l_2 = \theta_2 l_1$

30. An ideal gas exists in a state with pressure P_0, volume V_0.It is isothermally expanded to 4 times of its initial volume (V_0), then isobarically compressed to its original volume. Finally the system is heated isochorically to bring it to its initial state. The amount of heat exchanged in this process is :

(1) $P_0V_0(2\ln 2 - 0.75)$ (2) $P_0V_0(\ln 2 - 0.75)$

(3) $P_0V_0(\ln 2 - 0.25)$ (4) $P_0V_0(2\ln 2 - 0.25)$

31. A sinusoidal wave of wavelength 7.5 cm travels a distance of 1.2 cm along the x-direction in 0.3 sec. The crest P is at x = 0 at t = 0 sec and maximum displacement of the wave is 2 cm. Which equation correctly represents this wave ?

(1) $y = 2\cos(0.83x - 3.35t)$ cm

(2) $y = 2\sin(0.83x - 3.5t)$ cm

(3) $y = 2\cos(3.35x - 0.83t)$ cm

(4) $y = 2\cos(0.13x - 0.5t)$ cm

32. An electron is made to enters symmetrically between two parallel and equally but oppositely charged metal plates, each of 10 cm length. The electron emerges out of the field region with a horizontal component of velocity 10^6 m/s. If the magnitude of the electric between the plates is 9.1 V/cm, then the vertical component of velocity of electron is

(mass of electron = 9.1×10^{-31} kg and charge of electron = 1.6×10^{-19} C)

(1) 1×10^6 m/s (2) 0

(3) 16×10^6 m/s (4) 16×10^4 m/s

33. A coil of area A and N turns is rotating with angular velocity ω in a uniform magnetic field $\vec{B}$ about an axis perpendicular to $\vec{B}$. Magnetic flux φ and induced emf ε across it, at an instant when $\vec{B}$ is parallel to the plane of coil, are :

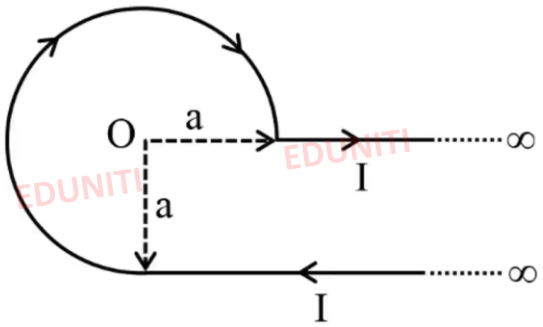
(1) $\varphi = AB, \varepsilon = 0$ (2) $\varphi = 0, \varepsilon = NAB\omega$

(3) $\varphi = 0, \varepsilon = 0$ (4) $\varphi = AB, \varepsilon = NAB\omega$

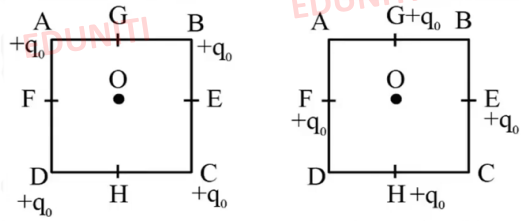
34. An infinite wire has a circular bend of radius a, and carrying a current I as shown in figure. The magnitude of magnetic field at the origin O of the arc is given by :

(1) $\frac{\mu_0 I}{4\pi a}\left[\frac{\pi}{2}+1\right]$ (2) $\frac{\mu_0 I}{4\pi a}\left[\frac{3\pi}{2}+1\right]$

(3) $\frac{\mu_0 I}{2\pi a}\left[\frac{\pi}{2}+2\right]$ (4) $\frac{\mu_0 I}{4\pi a}\left[\frac{3\pi}{2}+2\right]$



35. In the first configuration (1) as shown in the figure, four identical charges (q_0) are kept at the corners A,B,C and D of square of side length 'a'. In the second configuration (2), the same charges are shifted to mid points G,E,H and F, of the square, If $K = \frac{1}{4\pi\varepsilon_0}$, the difference between the potential energies of configuration (2) and (1) is given by :



(1) $\frac{Kq_0^2}{a}\left(4\sqrt{2}-2\right)$ (2) $\frac{Kq_0^2}{a}\left(3-\sqrt{2}\right)$

(3) $\frac{Kq_0^2}{a}\left(4-2\sqrt{2}\right)$ (4) $\frac{Kq_0^2}{a}\left(3\sqrt{2}-2\right)$

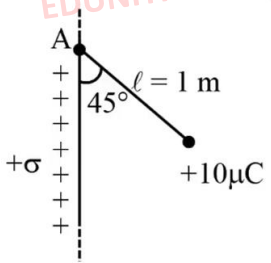
36. Consider a long straight wire of a circular cross-section (radius a) carrying a steady current I. The current is uniformly distributed across this cross-section. The distances from the centre of the wire's cross-section at which the magnetic field [inside the wire, outside the wire] is half of the maximum possible magnetic field, any where due to the wire, will be

(1) [a/4, 3a/2] (2) [a/2, 2a]

(3) [a/2, 3a] (4) [a/4, 2a]

37. A small bob of mass 100 mg and charge +10 μC is connected to an insulating string of length 1 m. It is brought near to an infinitely long non–conducting sheet of charge density 'σ' as shown in figure. If string subtends an angle of 45° with the sheet at equilibrium the charge density of sheet will be :

(Given, $\varepsilon_0 = 8.85 \times 10^{-12} \frac{F}{m}$ and acceleration due to gravity, $g = 10\ m/s^2$)



(1) 0.885 nC/m² (2) 17.7 nC/m²

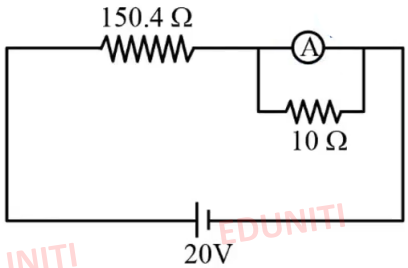
(3) 885 nC/m² (4) 1.77 nC/m²

38. A dipole with two electric charges of 2 μC magnitude each, with separation distance 0.5 μm, is placed between the plates of a capacitor such that its axis is parallel to an electric field established between the plates when a potential difference of 5 V is applied. Separation between the plates is 0.5 mm. If the dipole is rotated by 30° from the axis, it tends to realign in the direction due to a torque. The value of torque is :

(1) 5×10^{-9} Nm (2) 5×10^{-3} Nm

(2) 2.5×10^{-12} Nm (4) 2.5×10^{-9} Nm

39. In the figure shown below, a resistance of 150.4 Ω is connected in series to an ammeter A of resistance 240 Ω. A shunt resistance of 10 Ω is connected in parallel with the ammeter. The reading of the ammeter is _____ mA.



40. An infinitely long wire has uniform linear charge density $\lambda = 2$ nC/m. The net flux through a Gaussian cube of side length $\sqrt{3}$ cm, if the wire passes through any two corners of the cube, that are maximally displaced from each other, would be xNm^2C^{-1}, where x is :

[Neglect any edge effects and use $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ SI units]

(1) 0.72 π (2) 1.44 π

(3) 6.48 π (4) 2.16 π

41. Two cells of emf 1V and 2V and internal resistance 2 Ω and 1 Ω, respectively, are connected in series with an external resistance of 6 Ω. The total current in the circuit is I_1. Now the same two cells in parallel configuration are connected to same external resistance. In this case, the total current drawn is I_2. The value of $\left(\frac{I_1}{I_2}\right)$ is $\frac{x}{3}$. The value of x is _____.

42. A tightly wound long solenoid carries a current of 1.5 A. An electron is executing uniform circular motion inside the solenoid with a time period of 75ns. The number of turns per metre in the solenoid is_____.

[Take mass of electron $m_e = 9 \times 10^{-31}$ kg, charge of electron $|q_e| = 1.6 \times 10^{-19}$ C, $\mu_0 = 4\pi \times 10^{-7} \frac{N}{A^2}$, 1 ns $= 10^{-9}$ s]

43. In a moving coil galvanometer, two moving coils M_1 and M_2 have the following particulars :

$R_1 = 5\ \Omega$, $N_1 = 15$, $A_1 = 3.6 \times 10^{-3}\ m^2$, $B_1 = 0.25$ T

$R_2 = 7\ \Omega$, $N_2 = 21$, $A_2 = 1.8 \times 10^{-3}\ m^2$, $B_2 = 0.50$ T

Assuming that torsional constant of the springs are same for both coils, what will be the ratio of voltage sensitivity of M_1 and M_2 ?

(1) 1 : 1 (2) 1 : 4

(3) 1 : 3 (4) 1 : 2

44. The entropy of any system is given by

$$S = \alpha^2\beta \ln\left[\frac{\mu kR}{J\beta^2} + 3\right]$$

where α and β are the constants. μ, J, k and R are no. of moles, mechanical equivalent of heat, Boltzmann constant and gas constant respectively.

$$\left[\text{Take } S = \frac{dQ}{T}\right]$$

Choose the incorrect option from the following :

(1) α and J have the same dimensions.

(2) S, β, k and μR have the same dimensions.

(3) S and α have different dimensions.

(4) α and k have the same dimensions.

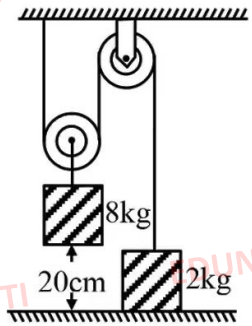
45. Water droplets are coming from an open tap at a particular rate. The spacing between a droplet observed at 4th second after its fall to the next droplet is 34.3 m. At what rate the droplets are coming from the tap ? (Take $g = 9.8\ m/s^2$)

(1) 3 drops / 2 seconds (3) 1 drop / second

(2) 2 drops / second (4) 1 drop / 7 seconds

CLICK FOR SOLUTION

46. The boxes of masses 2 kg and 8 kg are connected by a massless string passing over smooth pulleys. Calculate the time taken by box of mass 8 kg to strike the ground starting from rest. (use $g = 10\ m/s^2$)



(1) 0.34 s (2) 0.2 s

(3) 0.25 s (4) 0.4 s

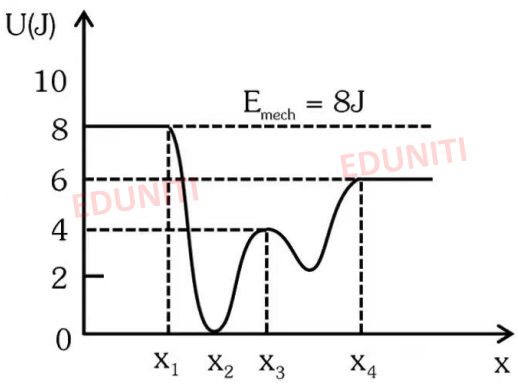
47. Given below is the plot of a potential energy function U(x) for a system, in which a particle is in one dimensional motion, while a conservative force F(x) acts on it. Suppose that $E_{mech} = 8$ J, the incorrect statement for this system is :

(1) at $x > x_4$, K.E. is constant throughout the region.

(2) at $x < x_1$, K.E. is smallest and the particle is moving at the slowest speed.

(3) at $x = x_2$, K.E. is greatest and the particle is moving at the fastest speed.

(4) at $x = x_3$, K.E. = 4 J.



48. A particle of mass 1 mg and charge q is lying at the mid-point of two stationary particles kept at a distance '2 m' when each is carrying same charge 'q'. If the free charged particle is displaced from its equilibrium position through distance 'x' ($x << 1$ m). The particle executes SHM. Its angular frequency of oscillation will be _____ $\times 10^5$ rad/s if $q^2 = 10\ C^2$.

49. Due to cold weather a 1 m water pipe of cross–sectional area 1 cm^2 is filled with ice at –10°C. Resistive heating is used to melt the ice. Current of 0.5 A is passed through 4 kΩ resistance. Assuming that all the heat produced is used for melting, what is the minimum time required ?

(Given latent heat of fusion for water/ice $= 3.33 \times 10^5\ J\ kg^{-1}$, specific heat of ice $= 2 \times 10^3\ J\ kg^{-1}$ and density of ice $= 10^3\ kg/m^3$

(1) 0.353 s (2) 35.3 s

(3) 3.53 s (4) 70.6 s

50. The image of an object placed in air formed by a convex refracting surface is at a distance of 10 m behind the surface. The image is real and is at $\frac{2^{rd}}{3}$ of the distance of the object from the surface .The wavelength of light inside the surface is $\frac{2}{3}$ times the wavelength in air. The radius of the curved surface is $\frac{x}{13}$ m. the value of 'x' is_____.

51. The electric field in an electromagnetic wave is given by $E = (50\ NC^{-1})\ \sin\omega\ (t - x/c)$

The energy contained in a cylinder of volume V is 5.5×10^{-12} J. The value of V is _____ cm^3.

$\left(\text{given} \in_0 = 8.8 \times 10^{-12} C^2 N^{-1} m^{-2}\right)$

CLICK FOR SOLUTION

52. A glass tumbler having inner depth of 17.5 cm is kept on a table. A student starts pouring water ($\mu = 4/3$) into it while looking at the surface of water from the above. When he feels that the tumbler is half filled, he stops pouring water. Up to what height, the tumbler is actually filled ?

(1) 11.7 cm

(2) 10 cm

(3) 7.5 cm

(4) 8.75 cm

53. Electric field of plane electromagnetic wave propagating through a non–magnetic medium is given by $E = 20\cos(2 \times 10^{10}\, t - 200x)$ V/m. The dielectric constant of the medium is equal to :

(Take $\mu_r = 1$)

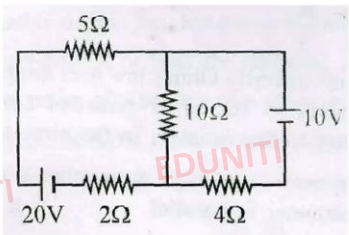
(1) 9 (2) 2 (3) $\frac{1}{3}$ (4) 3

54. Consider a uniform rod of mass $M = 4m$ and length l pivoted about its centre. A mass m moving with velocity v making angle $\theta = \pi/4$ to the rod's long axis collides with one end of the rod and sticks to it. The angular speed of the rod-mass system just after the collision is

(a) $\frac{4}{7}\frac{v}{l}$ (b) $\frac{3\sqrt{2}}{7}\frac{v}{l}$

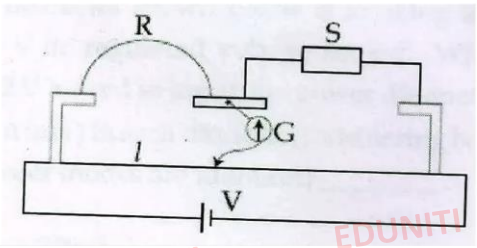
(c) $\frac{3}{7\sqrt{2}}\frac{v}{l}$ (d) $\frac{3}{7}\frac{v}{l}$

55. In the figure shown, the current in the 10V battery is close to

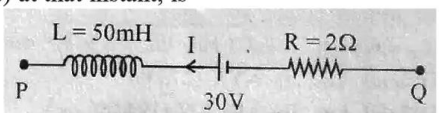


(a) 0.36 A from negative to positive terminal.
(b) 0.71 A from positive to negative terminal.
(c) 0.21 A from positive to negative terminal.
(d) 0.42 A from positive to negative terminal.

56. In a meter bridge experiment, S is a standard resistance. R is a resistance wire. It is found that balancing length is $l = 25$ cm. If R is replaced by a wire of half length and half diameter that of R of same material, then the balancing distance l' (in cm) will now be



57. A part of a complete circuit is shown in the figure. At some instant, the value of current I is 1 A and it is decreasing at a rate of 10^2 A s^{-1}. The value of the potential difference $V_P - V_Q$, (in volts) at that instant, is



58. An air bubble of volume 1cm^3 rises from the bottom of a lake 40m deep to the surface at a temperature of 12°C. The atmospheric pressure is 1×10^5 Pa, the density of water is 1000 kg/m^3 and g =10m/s^2. There is no difference of the temperature of water at the depth of 40m and on the surface. The volume of air bubble when it reaches the surface will be

(1) 5cm^3 (2) 2cm^3 (3) 4cm^3 (4) 3cm^3

59. 1 kg of water at 100°C is converted into steam at 100°C by boiling at atmospheric pressure. The volume of water changes from 1.00×10^{-3} m^3 as a liquid to 1.671 m^3 as steam. The change in internal energy of the system during the process will be

(Given latent heat of vaporisaiton = 2257 kJ/kg. Atmospheric pressure = 1×10^5 Pa)

(1) + 2090 kJ (2) – 2090 kJ

(3) – 2426 kJ (4) + 2476 kJ

CLICK FOR SOLUTION

60. A cubical volume is bounded by the surfaces $x = 0, x = a, y = 0, y = a, z = 0, z = a$. The electric field in the region is given by $\vec{E} = E_0 x \hat{i}$. Where $E_0 = 4 \times 10^4$ NC^{-1} m^{-1}. If a = 2 cm, the charge contained in the cubical volume is $Q \times 10^{-14}$C. The value of Q is____.

Take $\in_0 = 9 \times 10^{-12}$ C^2/Nm^2)

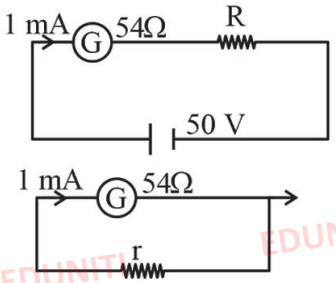
61. The mean free path of molecules of a certain gas at STP is 1500d, where d is the diameter of the gas molecules. While maintaining the standard pressure, the mean free path of the molecules at 373K is approximately :

(1) 1098d (2) 2049d

(3) 750d (4) 1500d

62. Two long parallel wires carrying currents 8A and 15 A in opposite directions are placed at a distance of 7 cm from each other. A point P is at equidistant from both the wires such that the lines joining the point P to the wires are perpendicular to each other. The magnitude of magnetic field at P is ____ $\times 10^{-6}$ T. (Given : $\sqrt{2} = 1.4$)

63. For designing a voltmeter of range 50 V and an ammeter of range 10 mA using a galvanometer which has a coil of resistance 54 Ω showing a full scale deflection for 1 mA as in figure.



(A) for voltmeter R ≈ 50 kΩ

(B) for ammeter r ≈ 0.2 Ω

(C) for ammeter r ≈ 6 Ω

(D) for voltmeter R ≈ 5 kΩ

(E) for voltmeter R ≈ 500 Ω

Choose the correct answer from the options given below :

(1) (C) and (E) (2) (C) and (D)

(3) (A) and (C) (4) (A) and (B)

64. The current sensitivity of moving coil galvanometer is increased by 25%. This increase is achieved only by changing in the number of turns of coils and area of cross section of the wire while keeping the resistance of galvanometer coil constant. The percentage change in the voltage sensitivity will be:

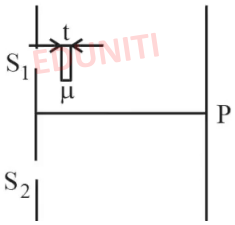
(1) +25% (2) – 50%

(3) Zero (4) – 25%

65. An ideal transformer with purely resistive load operates at 12 kV on the primary side. It supplies electrical energy to a number of nearby houses at 120 V. The average rate of energy consumption in the houses served by the transformer is 60 kW. The value of resistive load (Rs) required in the secondary circuit will be _____ mΩ.

CLICK FOR SOLUTION

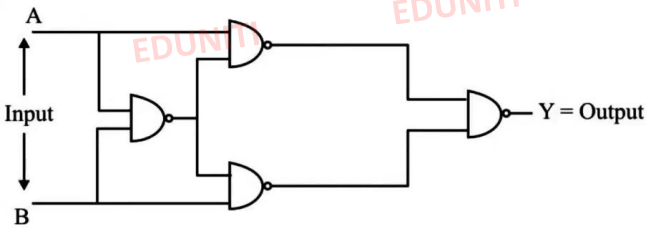
66. As shown in the figure, in Young's double slit experiment, a thin plate of thickness t = 10 μm and refractive index μ = 1.2 is inserted infront of slit S_1. The experiment is conducted in air (μ = 1) and uses a monochromatic light of wavelength λ = 500 nm. Due to the insertion of the plate, central maxima is shifted by a distance of $x\beta_0$. β_0 is the fringe-width before the insertion of the plate. The value of the x is_____.



67. In a Young's double slit experiment, the intensities at two points, for the path difference $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$ (λ being the wavelength of light used) are I_1 and I_2 respectively. If I_0 denotes the intensity produced by each one of the individual slits, then $\frac{I_1 + I_2}{I_0}$ = ...

68. Unpolarised light of intensity $32\,Wm^{-2}$ passes through the combination of three polaroids such that the pass axis of the last polaroid is perpendicular to that of the pass axis of first polaroid. If intensity of emerging light is $3\,Wm^{-2}$, then the angle between pass axis of first two polaroids is _____ ^{o}.

69. The output Y for the inputs A and B of circuit is given by



Truth table of the shown circuit is :

(1)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(2)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

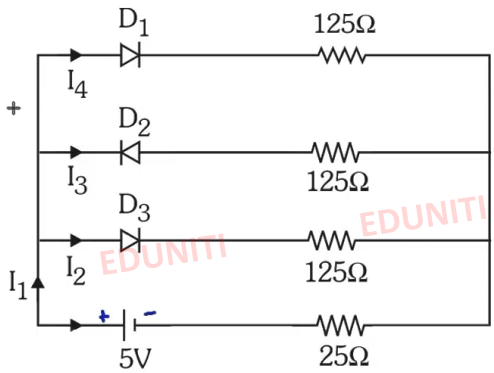
(3)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(4)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

70. If each diode has a forward bias resistance of 25 Ω in the below circuit,



Which of the following options is correct:

(1) $\frac{I_3}{I_4} = 1$

(2) $\frac{I_2}{I_3} = 1$

(3) $\frac{I_1}{I_2} = 1$

(4) $\frac{I_1}{I_2} = 2$

71. A monochromatic light is incident on a hydrogen sample in ground state. Hydrogen atoms absorb a fraction of light and subsequently emit radiation of six different wavelengths. The frequency of incident light is $x \times 10^{15}$ Hz. The value of x is _____. (Given $h = 4.25 \times 10^{-15}$ eVs)

CLICK FOR SOLUTION

72. A sphere of relative density σ and diameter D has concentric cavity of diameter d. The ratio of $\frac{D}{d}$, if it just floats on water in a tank is:

(1) $\left(\frac{\sigma}{\sigma-1}\right)^{\frac{1}{3}}$ (2) $\left(\frac{\sigma+1}{\sigma-1}\right)^{\frac{1}{3}}$

(3) $\left(\frac{\sigma-1}{\sigma}\right)^{\frac{1}{3}}$ (4) $\left(\frac{\sigma-2}{\sigma+2}\right)^{\frac{1}{3}}$

73. A parallel plate capacitor with plate separation 5 mm is charged up by a battery. It is found that on introducing a dielectric sheet of thickness 2 mm, while keeping the battery connections intact, the capacitor draws 25% more charge from the battery than before. The dielectric constant of the sheet is _____.

74. The effective resistance between A and B, if resistance of each resistor is R, will be



(1) $\frac{2}{3}R$ (2) $\frac{8R}{3}$

(3) $\frac{5R}{3}$ (4) $\frac{4R}{3}$

75. Identify the characteristics of an adiabatic process in a monoatomic gas.

(A) Internal energy is constant.

(B) Work done in the process is equal to the change in internal energy.

(C) The product of temperature and volume is a constant.

(D) The product of pressure and volume is a constant.

(E) The work done to change the temperature from T_1 to T_2 is proportional to $(T_2 - T_1)$

Choose the **correct** answer from the options given below :

(1) (A), (C), (D) only (2) (A), (C), (E) only

(3) (B), (E) only (4) (B), (D) only

CLICK FOR SOLUTION

EDUNITI

FILL YOUR ANSWER

Q	ANS
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Q	ANS
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Q	ANS
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74.	
75.	

CLICK FOR SOLUTION

ANSWER KEY

Q	ANS
1.	3
2.	4
3.	2
4.	2
5.	2
6.	3
7.	4
8.	4
9.	600
10.	6
11.	240
12.	C
13.	2
14.	15
15.	C
16.	44
17.	27
18.	B
19.	4
20.	2
21.	D
22.	1
23.	1
24.	4
25.	25
26.	2
27.	1
28.	4
29.	4
30.	1
31	1

Q	ANS
32.	3
33.	2
34.	2
35.	4
36.	2
37.	4
38.	1
39.	5
40.	4
41.	4
42.	250
43.	1
44.	4
45.	3
46.	4
47.	2
48.	6000
49.	2
50.	30
51.	500
52.	2
53.	1
54.	b
55.	c
56.	40
57.	33
58.	1
59.	1
60.	288
61.	2
62.	68

Q	ANS
63.	3
64.	1
65.	240
66.	4
67.	3
68.	30 or 60
69.	4
70.	4
71.	3
72.	1
73.	2
74.	2
75.	3

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